

Pizza Sales Analysis – SQL Case Study

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Business Problem & Objective

- This project analyses pizza sales data using SQL to evaluate business performance, identify key revenue drivers, and uncover customer ordering trends.
- The goal is to provide actionable insights to support better decision-making.





Dataset Description

This dataset contains transactional pizza sales data, including order details, product categories, sizes, and pricing. It is used to analyse revenue, sales trends, and customer ordering behaviour.

Total Revenue

SELECT

ROUND(SUM(total_price), 2) AS Total_Revenue

FROM

pizza_sales;

Result Grid	
	Total_Revenue
▶	817860.05

Average Order Value

```
SELECT
    ROUND(SUM(total_price) /
COUNT(DISTINCT order_id),2) AS Avg_Order_Value
FROM
    pizza_sales;
```

Result Grid	
	Avg_Order_Value
▶	38.31

Total Number of Orders

```
SELECT  
    COUNT(DISTINCT order_id) AS Total_Orders  
FROM  
    pizza_sales;
```

Result Grid	
	Total_Orders
▶	21350

Total Pizzas Sold

SELECT

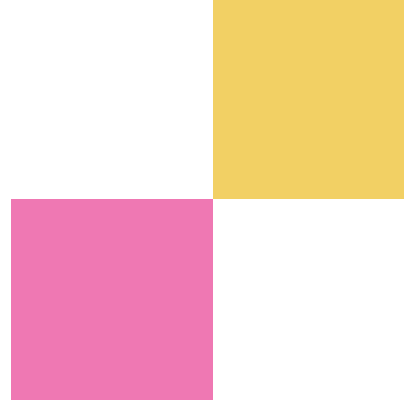
SUM(quantity) **AS** Total_Pizzas_Sold

FROM

pizza_sales;

Result Grid	
	Total_Pizzas_Sold
▶	49574

Total Pizzas Sold by Pizza Category



```
SELECT
    pizza_category, SUM(quantity) AS Total_Pizzas_Sold
FROM
    pizza_sales
GROUP BY pizza_category
ORDER BY Total_Pizzas_Sold DESC;
```

Result Grid			Filter Rows:
	pizza_category	Total_Pizzas_Sold	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	

Daily Order Trend

```
SELECT
```

```
    DAYNAME(order_date) AS Day_Name,
```

```
    COUNT(DISTINCT order_id) AS Total_Orders
```

```
FROM
```

```
    pizza_sales
```

```
GROUP BY Day_Name
```

```
ORDER BY Total_Orders DESC;
```

Result Grid   Filter Rows:		
	Day_Name	Total_Orders
▶	Friday	3538
	Thursday	3239
	Saturday	3158
	Wednesday	3024
	Tuesday	2973
	Monday	2794
	Sunday	2624

Monthly Order Trend

SELECT

MONTHNAME(order_date) **AS** Month_Name,
COUNT(**DISTINCT** order_id) **AS** Total_Orders

FROM

pizza_sales

GROUP BY Month_Name

ORDER BY Total_Orders **DESC**;

Result Grid   Filter Rows:		
	Month_Name	Total_Orders
▶	July	1935
	May	1853
	January	1845
	August	1841
	March	1840
	April	1799
	November	1792
	June	1773
	February	1685
	December	1680
	September	1661
	October	1646

Percentage Contribution of each Category to Total Revenue

SELECT

```
pizza_category,  
ROUND(SUM(total_price) , 2) AS Total_Sales,  
Concat(ROUND(SUM(total_price) * 100 /  
sum(sum(total_price)) over(), 2), '%')  
AS Percentage_of_Sales
```

FROM

pizza_sales

GROUP BY pizza_category;

Result Grid   Filter Rows: Ex			
	pizza_category	Total_Sales	Percentage_of_Sales
▶	Classic	220053.10	26.91%
	Veggie	193690.45	23.68%
	Supreme	208197.00	25.46%
	Chicken	195919.50	23.96%

Total 5 Pizzas by Revenue

```
SELECT
    pizza_name,
    ROUND(SUM(total_price), 2) AS Total_Revenue
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC
LIMIT 5;
```

Result Grid   Filter Rows:		
	pizza_name	Total_Revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768.00
	The California Chicken Pizza	41409.50
	The Classic Deluxe Pizza	38180.50
	The Spicy Italian Pizza	34831.25

Total 5 Pizzas by Quantity

SELECT

 pizza_name, SUM(quantity) **AS** Total_Quantity

FROM

 pizza_sales

GROUP BY pizza_name

ORDER BY Total_Quantity **DESC**

LIMIT 5;

Result Grid   Filter Rows:		
	pizza_name	Total_Quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Total 5 Pizzas by Total Orders

SELECT

 pizza_name,

 Count(**Distinct** order_id) **AS** Total_Orders

FROM

 pizza_sales

GROUP BY pizza_name

ORDER BY Total_Orders **Desc**

LIMIT 5;

Result Grid			Filter Rows:	
	pizza_name	Total_Orders		
▶	The Classic Deluxe Pizza	2329		
	The Hawaiian Pizza	2280		
	The Pepperoni Pizza	2278		
	The Barbecue Chicken Pizza	2273		
	The Thai Chicken Pizza	2225		

Bottom 5 Pizzas by Revenue

SELECT

 pizza_name, SUM(total_price) **AS** Total_Revenue

FROM

 pizza_sales

GROUP BY pizza_name

ORDER BY Total_Revenue **ASC**

LIMIT 5;

Result Grid |   Filter Rows:

	pizza_name	Total_Revenue
▶	The Brie Carre Pizza	11588.50
	The Green Garden Pizza	13955.75
	The Spinach Supreme Pizza	15277.75
	The Mediterranean Pizza	15360.50
	The Spinach Pesto Pizza	15596.00

Bottom 5 Pizzas by Quantity

```
SELECT
    pizza_name, SUM(quantity) AS Total_Quantity
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity Asc
LIMIT 5;
```

Result Grid   Filter Rows:		
	pizza_name	Total_Quantity
▶	The Brie Carre Pizza	490
	The Mediterranean Pizza	934
	The Calabrese Pizza	937
	The Spinach Supreme Pizza	950
	The Soppressata Pizza	961

Bottom 5 Pizzas by Total Orders

```
SELECT
    pizza_name,
    Count(Distinct order_id) AS Total_Orders
FROM
    pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders Asc
LIMIT 5;
```

Result Grid			Filter Rows:
	pizza_name	Total_Orders	
▶	The Brie Carre Pizza	480	
	The Mediterranean Pizza	912	
	The Calabrese Pizza	918	
	The Spinach Supreme Pizza	918	
	The Chicken Pesto Pizza	938	

Top 5 Orders with above Average Spending

```
with order_totals as (  
    select order_id, sum(total_price) as order_value  
    from pizza_sales  
    group by order_id)  
select order_id,  
       order_value  
from order_totals  
where order_value > (  
    select avg(order_value) from order_totals)  
order by order_value DESC  
Limit 5;
```

Result Grid			Filter Rows:	
	order_id	order_value		
▶	18845	444.20		
	10760	417.15		
	1096	285.15		
	6169	284.00		
	740	280.95		

Total 5 Pizzas by Total Orders

SELECT

pizza_name,

Count(**Distinct** order_id) **AS** Total_Orders

FROM

pizza_sales

GROUP BY pizza_name

ORDER BY Total_Orders **Desc**

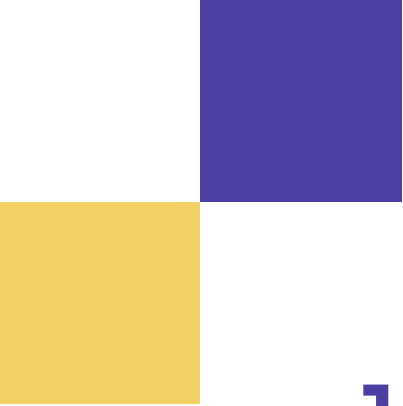
LIMIT 5;

Result Grid   Filter Rows:		
	pizza_name	Total_Orders
▶	The Classic Deluxe Pizza	2329
	The Hawaiian Pizza	2280
	The Pepperoni Pizza	2278
	The Barbecue Chicken Pizza	2273
	The Thai Chicken Pizza	2225

Most Popular Pizza Size per Category

```
SELECT *  
FROM (  
    SELECT pizza_category,  
           pizza_size,  
           SUM(quantity) AS total_qty,  
           dense_rank() OVER(PARTITION BY pizza_category  
                               ORDER BY SUM(quantity) DESC) AS rnk  
    FROM pizza_sales  
    GROUP BY pizza_category, pizza_size  
) t  
WHERE rnk = 1;
```

Result Grid				
Filter Rows:				
	pizza_category	pizza_size	total_qty	rnk
▶	Chicken	L	4932	1
	Classic	S	6139	1
	Supreme	L	4564	1
	Veggie	L	5403	1



KEY INSIGHTS

1. OVERALL BUSINESS PERFORMANCE

The business generated £800k revenue from 21k+ orders, showing consistent demand.

2. The Classic category generates the highest revenue (£220k), making it the top-performing segment.

3. Large-sized pizzas dominate sales across categories, indicating preference for group or value meals.

KEY INSIGHTS

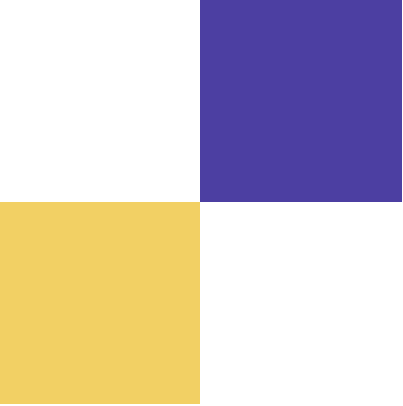
4. Friday records the highest number of orders, showing strong end-of-week demand.

5. Pizzas like Brie Carre generate very low revenue (~£11k), indicating poor performance.



Business Recommendations

1. Increase promotions for top-performing pizzas like Classic category and Thai Chicken Pizza, as they contribute the highest revenue.
2. Review low-performing items such as Brie Carre pizza, which contribute minimal revenue. Consider menu redesign, price adjustment, or removal to improve operational efficiency.
3. Introduce weekend combo deals to capitalise on peak demand.

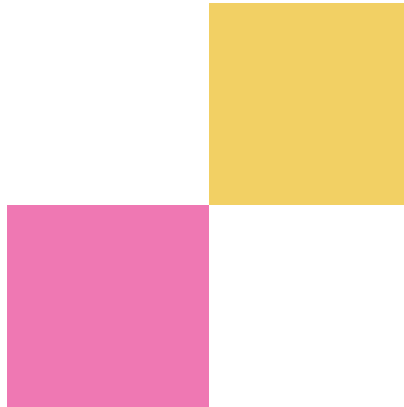


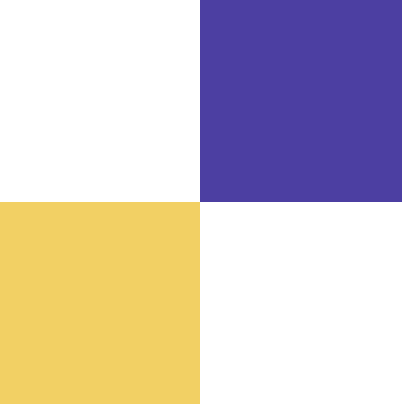
CONCLUSION

The analysis identified key revenue drivers, customer demand patterns, and product performance trends.

High performing categories like Classic pizzas and large sizes contribute significantly to overall sales.

Sales patterns show strong demand during weekends and peak periods.





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**This project demonstrates my ability to
transform raw data into actionable
insights to support data-driven business
decisions.**

